

Woojin Bae

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RESEARCH INTERESTS

Design and Fabrication of Nanomaterials through Advanced Techniques

- **Nanomaterials:** Perovskite, Zeolite, Metal Oxide, Nanoparticle.
- **Advanced Techniques:** Transmission Electron Microscopy, Autonomous Laboratories.

EDUCATION

Seoul National University (SNU), Seoul, South Korea Expected Aug. 2026

B.S. Chemical and Biological Engineering (CBE), *Cum Laude (expected)*

- GPA: 3.72/4.3 (overall), 3.67/4.0 (converted)
- Graduate Course: Topics in Inorganic Material and Semiconductor Processes (Fall 2025)

North Carolina State University (NCSU), Raleigh, NC May 2025

Exchange student in Chemical and Biomolecular Engineering

- GPA: 4.0/4.0, AIChE at NC State

Daejeon Science High School for the Gifted Feb. 2020

RESEARCH EXPERIENCE

Multi-dimensional Materials Chemistry Lab, SNU Mar. 2024 – Present

Undergraduate Researcher - (PI: *Prof. Jungwon Park*) Seoul, Korea

- **Team Leader;** Developed a DBSCAN-based protocol for quantitative analysis of SEI; SEI Structures Dependent Li-Morphology Characterized by cryo-TEM.
- Initiated and led a **collaborative project** across SNU and NCSU to characterize AI-driven synthesized Mn-doped CsPb(Cl/Br)₃ QDs, by TEM analysis and DFT calculation.
- Awarded a \$2,000 research grant by SNU College for a project across SNU and NCSU.

Summer Undergraduate Initiative (SURI), ASU May 2025 – Jul. 2025

Undergraduate Researcher - (PI: *Prof. Sandhya Susarla*) Tempe, AZ

- Selected for the **SURI fellowship** at Arizona State University (\$5000 stipend).
- Synthesized perovskite nanocrystals with controlled morphologies to investigate phonon–polariton coupling with hexagonal boron nitride (hBN).
- Presented a poster at Summer Research Initiative Final Research Showcase.

Self-Driving Fluidic Lab, NCSU Jan. 2025 – May 2025

Undergraduate Researcher - (PI: *Prof. Milad Abolhasani*) Raleigh, NC

- Achieved Photoluminescence Quantum Yield improvement of Mn-doped CsPb(Cl/Br)₃ perovskite through genetic algorithm-based synthesis optimization in autonomous lab.
- Presented a poster at Spring Undergraduate Research & Creativity Symposium.

PUBLICATIONS / PRESENTATIONS (†: equally contributed, *: corresponding author)

Jinge Xu.; **Bae, W.**; Milad Abolhasani. *, Autonomous Mn-doped Perovskite Nanocrystals synthesis by a Self-Driving lab. *Manuscript in preparation.*

Bae, W.†; Jung, S.†; Ji, S. †; Jeon, Y. *; Park, J. *, SEI Structures Dependent Li-Morphology Characterized by Cryo-TEM. *The 2024 Annual Fall Conference of Korean Society of Microscopy*, Gyeongju, Korea, Nov 25-26, 2024.

Bae, W.; Lee, K.; Lee, C.; Kim, T.; Choi, I. *, Regulation of astrocyte growth using porous titanium oxide shells. *Korean Chemical Society Future Chemist Research Presentation*, Daegu, Korea, Oct 19, 2018.

HONORS & AWARDS

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| Merit Award Korea Process Simulation Olympiad 2025, KIChE, Korea | Sep. 2025 |
| • Topic: Conducted feasibility study of Methanol synthesis process with Aveva process simulation (APS) (PI: <i>Prof. Jongmin Lee</i>) | |
| Excellence Research Award 2024 Student-Directed Research Symposium | Jan. 2025 |
| • Topic: SEI structures dependent Li-morphology characterized by Cryo-TEM. | |
| Korea-U.S. Exchange Scholarship Ministry of Trade, Industry and Energy, Korea | Nov. 2024 |
| • Awarded national \$9000 scholarship as a STEM-specialized exchange student. | |
| Excellence Award Undergraduate Research Program for Advanced Equipment | Jan. 2024 |
| • Awarded for excellence in undergraduate research on hollow ZSM-5 synthesis. | |
| Future Chemical Talent Award KUCST, Affiliation of Korean Chemical Society | Feb. 2020 |
| • Awarded as one of the top 6 science high school students in Korea. | |

LEADERSHIP & MEMBERSHIPS

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| Undergraduate Member Korean-American Scientists and Engineers Association | Jan. 2025 |
| • Delivered a research presentation at nationwide Korean-American undergraduate conference, Katalyst 2025. | |
| Tokyo Forum Youth Session The University of Tokyo, Japan | Fall 2024 |
| • Representative of Korea , in a session on Aging Society and Low Birth Rate. | |
| Young Engineers Honor Society National Academy of Engineering of Korea | Nov. 2024 |
| • Korean engineering honor society under National Academy of Engineering of Korea. | |
| Republic of Korea Air Force 11th Fighter Wing, Daegu, Korea | Aug. 2022 – May 2024 |
| Marketing team manager SNUfestival, Student Council, SNU | Jun. 2021 – May 2022 |
| • Promoted the festival through the online website and secured \$2200 sponsorship from a shared scooter company during the first in-person festival after the COVID-19 pandemic. | |
| Vice president Chemical and Biological Engineering Interim Leadership, SNU | Winter 2020 |
| • Developed an online program for incoming freshman during the COVID-19 pandemic. | |

TEACHING & MENTORING EXPERIENCES

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| Korean Language Peer Mentor Language Education Institute, SNU | Fall 2024 |
| Peer Tutor Basic Computing: First Adventures in Computing, SNU | Fall 2024, 2025 |
| Lecturer, TEM Seminar Multi-dimensional Materials Chemistry Lab, SNU | May 2024 |
| Student Mentor, SNU Mentoring SNU Social Responsibility | Apr. 2023 – Present |
| • Provided career guidance to underserved students, recognized as an outstanding mentor . | |

TECHNICAL SKILLS & LANGUAGES

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- **Instruments:** TEM, SEM, XRD, STEM/EDS, UV-Vis, OM, PL
 - **Programming Language:** Python, MATLAB, C
 - **Software:** Gatan DigitalMicrograph, VASP, APS, Adobe, Fusion 360
 - **Languages:** Korean (Native), English (C1, TOEFL 103)